Dolbeault moduli space of a stacky curve

Anna Borri

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1 Fundamental group

In this section, we recall the notion of topological Deligne-Mumford stack and its fundamental group, following [Noo00], [Noo05] and [Noo04].

1.1 Topological DM stacks

We follow [Noo05] in recalling the basic notions regarding topological stacks. Let **Top** be the category of compactly generated topological spaces. It becomes a site by taking the usual open coverings of topological spaces as coverings, therefore we can consider stacks over **Top**.

The usual notions related to stacks can be translated from the algebraic setting to the topological setting, e.g. representability and atlases.

Definition 1.1. [Noo05, Definition 7.1] An *atlas* of a stack $\mathcal{X}$ is a representable epimorphism $p\colon X \to \mathcal{X}$. If $\mathcal{X}$ admits such an atlas, it is called a *pretopological stack*.

Definition 1.2. [Noo05, Definition 13.8] A pretopological stack is called a *topological* or *weak Deligne-Mumford* if it admits a chart $p\colon X \to \mathcal{X}$ that is a local homeomorphism (namely, it pullbacks to surjective local homeomorphisms along topological spaces).

Definition 1.3. [Noo05, Definition 14.3] A *Deligne-Mumford topological stack* (DM topological stack) is a weak Deligne-Mumford topological stack that is locally a quotient by a *mild* action of the inertia groups.
Explicitly, it is required that for any point $x\colon * \to \mathcal{X}$ and for any open open substack $\mathcal{U} \subseteq \mathcal{X}$ containing x, there exists an open substack $\mathcal{V} \subseteq \mathcal{X}$ containing x such that $\mathcal{V} \cong [V/I_x]$, where V is a topological space on which the inertia group I_x acts mildly at x. By *mild* action at x we mean that any neighborhood of x in V contains an I_x-invariant neighborhood.

1.2 Homotopy theory and fundamental group

In this setting, it makes sense to introduce a notion of homotopy theory, that will be used to define the fundamental group.

As in classical topology, we can introduce the notion of *pair* of topological stacks $(\mathcal{X}, \mathcal{A})$ and of morphism of pairs $f\colon (\mathcal{X}, \mathcal{A}) \to (\mathcal{Y}, \mathcal{B})$, see [Noo05, Definition 17.1].

Definition 1.4. [Noo05, Definition 17.2] Let $f, g\colon (\mathcal{X}, \mathcal{A}) \to (\mathcal{Y}, \mathcal{B})$ be morphisms of pairs. A *homotpoy* from f to g is a triple $(H, \varepsilon_0, \varepsilon_1)$, where

- $H\colon ([0,1] \times \mathcal{X},\ [0,1] \times \mathcal{A}) \to (\mathcal{Y}, \mathcal{B})$ is a morphism of pairs;
- $\varepsilon_0\colon f \Rightarrow H_0$ and $\varepsilon_1\colon H_1 \Rightarrow g$ are natural isomorphisms.

A homotopy between points $x, y\colon * \to \mathcal{X}$ is called a *path*.

Remark 1.5. [Noo05, Remark 17.3] Note that any 2-isomorphism $f \cong g$ can be regarded as an homotopy.

With the notion of homotopy at hand, we can define the homotopy groups.

Definition 1.6. [Noo05, Definition 17.5] The *n-th homotopy group* of $\mathcal{X}$ pointed at $x \in \mathcal{X}$ is the set of pointed homotopy classes of morphisms from the n-sphere $\mathbb{S}^n$ into $\mathcal{X}$. Namely,

$$\pi_n(\mathcal{X}, x) \coloneqq [(\mathbb{S}^n, *), (\mathcal{X}, x)].$$

As in the classical topological case, the homotopy groups have a natural group structure and are functorial in X.

Continuing the analogy with classical topology and algebraic topology, we want to define a notion of covering space and observe that, under suitable topological hypotheses, the fundamental group $\pi_1(\mathcal{X}, x)$ can be recovered by the covering spaces of $\mathcal{X}$.

Definition 1.7. [Noo05, Definition 18.10] A morphism $\mathcal{Y} \to \mathcal{X}$ is a *covering map* if it is representable and for any morphism $W \to \mathcal{X}$ (with W topological space), the base change $W \times_{\mathcal{X}} \mathcal{Y} \to W$ is a covering map.

In order to recover the fundamental group from the covering spaces of $\mathcal{X}$, we need to define the notion of fiber and the fiber functor.

Definition 1.8. [Noo05, Definition 18.13] Given a morphism $f\colon \mathcal{Y} \to \mathcal{X}$, the *fiber* of f over $x \in \mathcal{X}$ is defined as the groupoid of the points of the pullback $* \times_{\mathcal{X}} \mathcal{Y}$ and denoted by $\mathcal{Y}_x$.

Assume that $\mathcal{X}$ is a path-connected topological stack.

Definition 1.9. [Noo05, Definition 18.13] Let $\mathbf{Cov}_{\mathcal{X}}$ be the category of covering spaces of $\mathcal{X}$. Define the fiber functor as

$$\begin{aligned} F_x\colon \mathbf{Cov}_{\mathcal{X}} &\to \pi_1(\mathcal{X}, x)\text{-}\mathbf{Set} \\ \mathcal{Y} &\mapsto \pi_0(\mathcal{Y}_x), \end{aligned}$$

where $\pi_0(Y_x)$ denotes the set of isomorphism classes of $\mathcal{Y}_x$, and the action of $\pi_1(\mathcal{X}, x)$ on $\pi_0(\mathcal{Y}_x)$ is an analogous to the monodromy action.

Theorem 1.10. [Noo05, Theorem 18.19] If $\mathcal{X}$ is connected, locally path connected and semilocally 1-connected, then the fiber functor F_x is an equivalence of categories.

Corollary 1.11. [Noo05, Corollary 18.20] Under the same hypotheses, there is a natural bijection between the subgroups $H \subseteq \pi_1(\mathcal{X}, x)$ (up to conjugation) and pointed connected coverings $p\colon (\mathcal{Y}, y) \to (\mathcal{X}, x)$ (up to isomorphsim). In particular, $\mathcal{X}$ has a *universal cover*.

Remark 1.12. The one presented is an analogous of Grothendieck's theory of Galois categories, presented in [Gro71]. The only difference is that in the classical theory of Galois categories the fiber functor is supposed to have values in finite sets. However, the results presented in [Gro71] can be easily generalized to this setting. Following this idea, Theorem 1.10 can be rephrased as follows.

Let $F'_x\colon \mathbf{Cov}_{\mathcal{X}} \to \mathbf{Set}$ be the fiber functor, considered as a $\mathbf{Set}$-valued functor. In analogy with the theory of Galois categories, we can define

$$\pi'_1(\mathcal{X}, x) \coloneqq \mathrm{Aut}(F'_x)$$

and prove that $\mathbf{Cov}_{\mathcal{X}}$ is equivalent to $\pi'_1(\mathcal{X}, x)$-$\mathbf{Set}$, see [Noo05, Remark 18.22]. If $\mathcal{X}$ is connected, locally path connected and semilocally 1-connected, using Theorem 1.10 we get an equivalence

$$\pi_1(\mathcal{X}, x) \cong \pi'_1(\mathcal{X}, x).$$

Just like in classical topology, this shows that, under these hypotheses, the fundamental group of $\mathcal{X}$ can be computed from the category of covering spaces of $\mathcal{X}$ and in particular it can be identified with the group of deck transformations of the universal cover of $\mathcal{X}$.

1.3 Seifert-Van Kampen theorem

Using the equivalence

$$\pi_1(\mathcal{X}, x) \cong \pi'_1(\mathcal{X}, x),$$

we can prove a version of the Seifert-Van Kampen theorem for topological stacks. This is analogous to the classical setting of Galois categories, as exposed for example in [DD20, Theorem 4.7.7]. We report the proof for completeness.

Theorem 1.13. Let $\mathcal{U}_1 \overset{i_1}{\hookrightarrow} \mathcal{X}$ and $\mathcal{U}_2 \overset{i_2}{\hookrightarrow} \mathcal{X}$ be open subschemes of $\mathcal{X}$ such that $\mathcal{X} = \mathcal{U}_1 \cup \mathcal{U}_2$. Let $\mathcal{V} \coloneqq \mathcal{U}_1 \times_{\mathcal{X}} \mathcal{U}_2$. Suppose that $\mathcal{U}_1, \mathcal{U}_2$ and $\mathcal{V}$ are connected. Then, for $x \in \mathcal{V}$, there is an isomorphism

$$\pi_1(\mathcal{X}, x) \cong \pi_1(\mathcal{U}_1, x) *_{\pi_1(\mathcal{V}, x)} \pi_1(\mathcal{U}_2, x).$$

Equivalently, the following diagram is cocartesian

$$\begin{tikzcd}
\pi_1(\mathcal{V}, x) \arrow[r] \arrow[d] & \pi_1(\mathcal{U}_1, x) \arrow[d] \\
\pi_1(\mathcal{U}_2, x) \arrow[r] \arrow[ur, phantom, "\ulcorner"] & \pi_1(\mathcal{X}, x).
\end{tikzcd}$$

Proof. In order to simplify the notation, let $G \coloneqq \pi_1(\mathcal{X}, x)$, $H \coloneqq \pi_1(\mathcal{V}, x)$ and $G_l \coloneqq \pi_1(\mathcal{U}_l, x)$ for $l = 1, 2,$. The inclusions

$$\begin{tikzcd} \mathcal{V} \arrow[r, hook, "j_l"] \arrow[rr, hook, bend left, "j"] & \mathcal{U}_l \arrow[r, hook, "i_l"] & \mathcal{X} \end{tikzcd}$$

give rise to a commutative diagram

$$\begin{tikzcd} H \arrow[r, "j_{1*}"] \arrow[d, "j_{2*}"'] & G_1 \arrow[d, "\iota_1"] \arrow[ddr, bend left, "i_{1*}"] & \\ G_2 \arrow[r, "\iota_2"] \arrow[drr, bend right, "i_{2*}"'] & G_1 *_H G_2 \arrow[dr, dashed, "v"] & \\ & & G, \end{tikzcd}$$

where the dashed arrow can be filled out by the universal property of $G_1 *_H G_2$. We want to prove that v is an isomorphism.

Consider the following 2-commutative diagram

$$\begin{tikzcd} \mathbf{Cov}_{\mathcal{X}} \arrow[rr, "\theta = F_x^{\mathcal{X}}"] \arrow[d, "\alpha"] & & G\text{-}\mathbf{Set} \arrow[d, "\beta"] \\ \mathbf{Cov}_{\mathcal{U}_1} \times_{\mathbf{Cov}_{\mathcal{V}}} \mathbf{Cov}_{\mathcal{U}_2} \arrow[rr, "\varphi = F_x^{\mathcal{U}_1} \times_{F_x^{\mathcal{V}}} F_x^{\mathcal{U}_2}"] & & G_1\text{-}\mathbf{Set} \times_{H\text{-}\mathbf{Set}} G_2\text{-}\mathbf{Set}, \end{tikzcd}$$

where α is induced by the restriction of covering spaces and β is induced by i_{1*} and i_{2*}. Notice that α is an equivalence of categories, obtained by the glueing of covering spaces, φ and θ are equivalences, as a consequence of Theorem 1.10. Therefore, β is an equivalence as well.

Notice moreover that the homomorphisms (ι_1, ι_2) induce an equivalence of categories

$$\gamma \colon (G_1 *_H G_2)\text{-}\mathbf{Set} \to G_1\text{-}\mathbf{Set} \times_{H\text{-}\mathbf{Set}} G_2\text{-}\mathbf{Set},$$

making the following diagram 2-commute

$$\begin{tikzcd} G\text{-}\mathbf{Set} \arrow[dr, "v^*"] \arrow[dd, "\beta"'] & \\ & (G_1 *_H G_2)\text{-}\mathbf{Set} \arrow[dl, "\gamma"] \\ G_1\text{-}\mathbf{Set} \times_{H-set} G_2\text{-}\mathbf{Set}. & \end{tikzcd}$$

Since both β and γ are equivalences, this shows that v^* is also an equivalence and therefore, see [DD20, Proposition 4.7.1], that v is an isomorphism of groups. $\square$

1.4 The case of a stacky curve

In this last paragraph, we want to apply the theory exposed earlier to an explicit example. The case we are interested in is the one of a *stacky curve*, more specifically a curve with a unique stacky point.

Let X be a smooth, projective, genus g curve over $\mathbb{C}$ and let $p \in X$ be a closed point. Following [Cad07], we construct $\mathcal{X} = \sqrt[e]{p/X}$ as a root stack over X, with multiplicity e in p.

Indeed, consider p as an effective Cartier divisor on X. Associated to it, there is a pair $(\mathcal{O}(p), s_p)$, where $\mathcal{O}(p)$ is a line bundle and s_p is a section of $\mathcal{O}(p)$ vanishing at p, called the *tautological section*. This pair defines a morphism

$$X \xrightarrow{(\mathcal{O}(p),s_p)} [\mathbb{A}^1/\mathbb{G}_m].$$

Let $\wedge e\colon [\mathbb{A}^1/\mathbb{G}_m] \to [\mathbb{A}^1/\mathbb{G}_m]$ be the morphism induced by taking the e-th power maps on $\mathbb{A}^1$ and on $\mathbb{G}_m$. Define $\mathcal{X}$ as the pullback

$$\begin{tikzcd} \mathcal{X} \arrow[r] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner", very near start] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge e"] \\ X \arrow[r, "(\mathcal{O}(p),s_p)"'] & {[\mathbb{A}^1/\mathbb{G}_m]}. \end{tikzcd} \tag{1.1}$$

The vertical map π is a coarse moduli map, see [Cad07, Corollary 2.3.7]. The stack $\mathcal{X}$ comes equipped with a pair $(\mathcal{O}(\frac{1}{e}p), \frac{1}{e}s_p)$, where $\mathcal{O}(\frac{1}{e}p)$ is a line bundle and $\frac{1}{e}s_p$ is a section of $\mathcal{O}(\frac{1}{e}p)$. This pair corresponds to the upper horizontal morphism in (1.1), and is characterized by the properties

$$\pi^*\mathcal{O}(p) = \mathcal{O}(\tfrac{1}{e}p)^{\otimes e}, \quad \pi^* s_p = (\tfrac{1}{e}s_p)^{\otimes e}.$$

The sheaf $\mathcal{O}(\frac{1}{e}p)$ is called the *tautological sheaf* of the stacky curve $\mathcal{X}$. It corresponds to the composition

$$\mathcal{X} \to [\mathbb{A}^1/\mathbb{G}_m] \to B\mathbb{G}_m.$$

With an abuse of notation, we denote by p also the stacky point in $\mathcal{X}$ above p. Notice that in p the scheme $\mathcal{X}$ has the structure

$$\begin{tikzcd} B\mu_e \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{X} \arrow[d, "\pi"] \\ \{p\} \arrow[r, hook] & X. \end{tikzcd} \tag{1.2}$$

More explicitly, if $U \cong \mathbb{A}^1$ is an affine neighborhood of p, we have that

$$\mathcal{X}|_U \cong [\mathbb{A}^1/\mu_e], \tag{1.3}$$

see [Cad07, Lemma 2.3.1]. Away from p the morphism π is an isomorphism

$$\mathcal{X} \setminus \{p\} \cong X \setminus \{p\}. \tag{1.4}$$

Therefore, we can use the isomorphisms (1.3) and (1.4) to compute the fundamental group of $\mathcal{X}$. Let U be as above, consider $\mathcal{U}_1 = \pi^{-1}(U), \mathcal{U}_2 = \mathcal{X}\setminus\{p\}$ and take a point $x \in \mathcal{U}_1 \cap \mathcal{U}_2 =: \mathcal{V}$. Then by (1.4) the substack $\mathcal{U}_2$ is a punctured curve and therefore

$$\pi_1(\mathcal{U}_2, x) = \langle a_i, b_i \mid i = 1, \dots g\rangle.$$

In order to compute $\pi_1(\mathcal{U}_1, x)$, notice that $[\mathbb{A}^1/\mu_e]$ is a Deligne-Mumford stack with atlas $q\colon \mathbb{A}^1 \to [\mathbb{A}^1/\mu_e]$. In particular, the morphism q is étale, i.e. a covering map. Since $\mathbb{A}^1$ is simply connected, this is the universal cover of $\mathcal{U}_1$. As observed in the previous paragraph, the fundamental group can be identified with the group of deck transformations of the universal cover. Since q is a μ_e-bundle, this shows that

$$\pi_1(\mathcal{U}_1, x) = \mu_e = \langle c \mid c^e = 1\rangle.$$

Finally, the intersection $\mathcal{V}$ is isomorphic to $\mathbb{A}^1 \setminus \{0\}$, therefore

$$\pi_1(\mathcal{V}, x) = \langle d\rangle.$$

Now we can use Theorem 1.13 to compute the fundamental group of $\mathcal{X}$. Indeed, the inclusions $j_1\colon \mathcal{V} \to \mathcal{U}_1$ and $j_2\colon \mathcal{V} \to \mathcal{U}_2$ induce

$$\begin{aligned} j_{1*}\colon \pi_1(\mathcal{V}, x) &\to \pi_1(\mathcal{U}_1, x) \\ d &\mapsto c, \end{aligned}$$

$$\begin{aligned} j_{2*}\colon \pi_1(\mathcal{V}, x) &\to \pi_1(\mathcal{U}_2, x) \\ d &\mapsto \prod_{i=1}^{g} [a_i, b_i]. \end{aligned}$$

Therefore we get the following.

Proposition 1.14. The fundamental group of the stacky curve $\mathcal{X}$ defined by Equation (1.1) is

$$\pi_1(\mathcal{X}) = \left\langle a_i,\, b_i,\, c \;\middle|\; \prod_{i=1}^{g} [a_i, b_i] = c,\; c^e = 1 \right\rangle. \tag{1.5}$$

2 Sheaves on $[\operatorname{Spec} A/G]$

In the previous section, we introduced a stacky curve $\mathcal{X}$ and studied its fundamental group. In what follows, we want to study different moduli spaces of bundles on $\mathcal{X}$ and understand how they are related to the corresponding moduli spaces on X. As observed in (1.3), the local structure of $\mathcal{X}$ around the stacky point is $[\mathbb{A}^1/\mu_e]$. Hence, a first step for understanding bundles on $\mathcal{X}$ is to understand coherent sheaves on $[\mathbb{A}^1/\mu_e]$. In this section, we study more in general coherent sheaves on quotient stacks of the form $[\operatorname{Spec} A/G]$, where G is a discrete group acting on $\operatorname{Spec} A$.

In the rest of the section, we denote $\operatorname{Spec} A$ by X and we work over $\operatorname{Spec} k$, where k is an algebraically closed field of characteristic 0.

2.1 Coherent sheaves

The quotient map $X \to [X/G]$ is an atlas. Therefore, the datum of coherent sheaf on $[X/G]$ is equivalent to the datum of a coherent sheaf $\mathcal{F}$ on X with descent data. The descent data is composed by an isomorphism

$$\alpha\colon pr_1^*\mathcal{F} \to pr_2^*\mathcal{F}$$

of coherent sheaves on $X \times_{BG} X$, and by natural isomorphisms on $X \times_{BG} X \times_{BG} X$, giving the cocycle condition.

Remark 2.1. There is an isomorphism $X \times_{BG} X \cong G \times X$. Composing with this isomorphism, the projection $pr_2\colon X \times_{BG} X \to X$ is the natural projection $G \times X \to X$ and the projection $pr_1\colon X \times_{BG} X \to X$ is the group action $\sigma\colon G \times X \to X$.

In the case $X = \operatorname{Spec} A$, the coherent sheaf $\mathcal{F}$ corresponds to a finitely presented A-module M. The isomorphism of $\mathcal{O}_{G\times X}$-modules

$$\alpha\colon pr_2^*\mathcal{F} \to \sigma^*\mathcal{F}$$

corresponds to an isomorphism of $(k[G] \otimes A)$-modules

$$\alpha^{\#}\colon k[G] \otimes M \to k[G] \otimes M,$$

where on the left hand side we have the natural structure of $k[G] \otimes A$-module, while on the right hand side $k[G]$ acts also on M via the action on A. This gives a morphism of A-modules

$$\rho\colon M \to k[G] \otimes M.$$

The morphism ρ gives M the structure od a $k[G]$-comodule. Indeed, the commutativity of

$$\begin{tikzcd}
k[G] \arrow[r, "\rho"] \arrow[d, "\rho"] & k[G] \otimes M \arrow[d, "\mathrm{id}_{k[G]} \otimes \rho"] \\
k[G] \otimes M \arrow[r, "\Delta \otimes \mathrm{id}_M"'] & k[G] \otimes k[G] \otimes M
\end{tikzcd}$$

follows from the cocycle condition expressed by

$$\begin{tikzcd}
X \times_{[X/G]} X \times_{[X/G]} X \arrow[r, equal] & G \times G \times X \arrow[r, "\mathrm{id}_G \times \sigma"] \arrow[r, bend left, "m \times \mathrm{id}_X"] \arrow[r, bend right, "pr_{13}"'] & G \times X \arrow[r, shift left, "\rho"] \arrow[r, shift right, "pr_2"'] & X.
\end{tikzcd}$$

While the commutativity of

$$\begin{tikzcd}
M \arrow[r, "\rho"] \arrow[dr, bend right, "\mathrm{id}_M"'] & k[G] \otimes M \arrow[d, "\varepsilon \otimes \mathrm{id}_M"] \\
& M \cong k \otimes M
\end{tikzcd}$$

follows from the fact that $\alpha^\#$ is an isomorphism of $k[G]$-modules.

In conclusion, the datum of a coherent sheaf on $[\operatorname{Spec} A/G]$ is equivalent to the datum of an A-module M with the structure of $k[G]$-comodule. But to give M the structure of $k[G]$-comodule is equivalent to giving it the structure of a G-representation. Therefore, coherent sheaves on $[\operatorname{Spec} A/G]$ are A-modules with a G-action. In the case $A = k$, we get an equivalence

$$\langle \text{Coherent sheaves on } BG \rangle \cong \langle \text{Finite dimensional } G\text{-representations} \rangle .$$

2.2 Vector bundles

As just mentioned, a coherent sheaf on BG gives a G-representation. Let us consider more in detail the case of vector bundles. A vector bundle $\mathcal{E}$ on BG corresponds to a morphism $BG \to BGL_n$.

Remark 2.2. A morphism $f\colon BG \to BGL_n$ yields a homomorphism of group schemes $G \to GL_n$.

Proof. For any test scheme T, consider the trivial G-torsor $G \times T \to T$ as an object of $BG(T)$. Then

$$\operatorname{Aut}_{BG(T)}(G \times T \to T) = G(T).$$

Notice that f must map trivial torsors to trivial torsors. Indeed, for $T = \operatorname{Spec} k$ we have that $f(G \to \operatorname{Spec} k)$ is a GL_n-torsor over $\operatorname{Spec} k$. Since k is algebraically closed, $\operatorname{Spec} k$ admits only trivial torsors, therefore $f(G \to \operatorname{Spec} k)$ is the trivial torsor $GL_n \to \operatorname{Spec} k$. For other schemes $T \in \operatorname{Sch}/k$ the statement follows by compatibility with pullback.
Passing to the automorphisms groupoids, we get that f induces a morphism of groupoids

$$G(T) = \operatorname{Aut}_{BG(T)}(G \times T \to T) \to \operatorname{Aut}_{BGL_n}(GL_n \times T \to T) = GL_n(T),$$

which is functorial in T. This is exactly a morphism of group schemes $G \to GL_n$, i.e. an n-dimensional representation of G. $\square$

Remark 2.3. The G-representation $G \to GL_n$ found above is the same as the representation obtained by the descent data of the vector bundle $\mathcal{E}$ corresponding to the morphism $BG \to BGL_n$, as mentioned in Section 2.1.

2.3 The case of a stacky curve

We now apply the theory to the example presented in Section 1.4. As already mentioned, $\mathcal{X}$ is equipped with a line bundle $\mathcal{O}(\frac{1}{e}p)$.

We are interested in studying the behavior of $\mathcal{O}(\frac{1}{e}p)$ at the stacky point of $\mathcal{X}$. Namely, considering the fiber product

$$\begin{tikzcd} B\mu_e \arrow[r, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "\pi"] \\ * \arrow[r, "p"] & X, \end{tikzcd}$$

we are interested in the pullback $i^*\mathcal{O}(\frac{1}{e}p)$, which corresponds to the composition

$$B\mu_n \xrightarrow{i} \mathcal{X} \to [\mathbb{A}^1/\mathbb{G}_m] \to B\mathbb{G}_m.$$

Theorem 2.4. [Taa24, Theorem 1.1.36] In the following diagram, the outer square is cartesian

$$\begin{tikzcd}
B\mu_e \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & \mathcal{X} \arrow[r] \arrow[d, "\pi"] \arrow[dr, phantom, "\lrcorner", very near start] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[r] \arrow[d, "\wedge e"] & B\mathbb{G}_m \arrow[d, "\wedge e"] \\
* \arrow[r, "p"'] & X \arrow[r] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[r] & B\mathbb{G}_m.
\end{tikzcd}$$

In particular, it is the pullback square

$$\begin{tikzcd}
B\mu_e \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner", very near start] & B\mathbb{G}_m \arrow[d, "\wedge e"] \\
* \arrow[r] & B\mathbb{G}_m
\end{tikzcd}$$

corresponding to the exact sequence of groups

$$1 \to \mu_e \to \mathbb{G}_m \xrightarrow{\wedge e} \mathbb{G}_m.$$

Proof. Consider the left side of the diagram

$$\begin{tikzcd}
B\mu_e \arrow[r, "(\pi^*\mathcal{O}(p)|_{B\mu_e},\, \pi^* s_p|_{B\mu_e})"] \arrow[d, "\psi"] \arrow[dr, phantom, "\lrcorner", very near start] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge e"] \\
* \arrow[r, "(\mathcal{O}(p)|_p,\, s_p|_p)"'] & {[\mathbb{A}^1/\mathbb{G}_m]}.
\end{tikzcd} \tag{2.1}$$

Since the section s_p by definition vanishes at p, we have $s_p|_p = 0$. Moreover, by construction

$$(\pi^* s_p|_{B\mu_e})^e = \psi^* s_p|_p = \psi^* 0 = 0,$$

therefore $\pi^* s_p|_{B\mu_e} = 0$. This shows that, in the diagram (2.1), the horizontal arrows factor through the inclusion $B\mathbb{G}_m \to [\mathbb{A}^1/\mathbb{G}_m]$ given by the choice of the zero section. So we get

$$\begin{tikzcd}
B\mu_e \arrow[r, "\mathcal{O}(\frac{1}{e}p)|_{B\mu_e}"] \arrow[d] & B\mathbb{G}_m \arrow[r] \arrow[d, dashed] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge e"] \\
* \arrow[r, "\mathcal{O}(p)|_p"] & B\mathbb{G}_m \arrow[r] & {[\mathbb{A}^1/\mathbb{G}_m]},
\end{tikzcd} \tag{2.2}$$

where the vertical arrow can be filled by the map induced by the e-th power morphism $\mathbb{G}_m \to \mathbb{G}_m$. Then, the right square in (2.2) becomes the cartesian

square

$$\begin{tikzcd} B\mathbb{G}_m \arrow[r] \arrow[d, "\wedge e"] \arrow[dr, phantom, "\lrcorner"] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge e"] \\ B\mathbb{G}_m \arrow[r] & {[\mathbb{A}^1/\mathbb{G}_m]}. \end{tikzcd}$$

This shows that in (2.2) both the outer and the rightmost squares are cartesian. Therefore, the leftmost square is also cartesian. This means that $\mathcal{O}(\frac{1}{e}p)|_{B\mu_e}$ is the line bundle obtained by the pullback square

$$\begin{tikzcd} B\mu_e \arrow[r] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & B\mathbb{G}_m \arrow[d, "\wedge e"] \\ * \arrow[r] & B\mathbb{G}_m, \end{tikzcd}$$

corresponding to the inclusion $\mu_e \hookrightarrow \mathbb{G}_m$. □

Remark 2.5. The one above is a modified version of the proof given in [Taa24], which we think might provide more clarity.

This means that the μ_e-representation obtained as $\mathcal{O}(\frac{1}{e}p)|_{B\mu_e}$ is just the tautological character, corresponding to the inclusion $\mu_e \hookrightarrow \mathbb{G}_m$.

3 Sheaves on a stacky curve

In Section 2, we studied the behavior of the tautological sheaf $\mathcal{O}(\frac{1}{e}p)$ at the stacky point of the root stack $\mathcal{X}$. In this section we move on to further understand vector bundles on $\mathcal{X}$. We follow [Taa24], and work in a slightly more general setting than before, namely we consider $\mathcal{C}$ a stacky curve with possibly finitely many stacky points and C its coarse moduli space. In the last section, we will apply the theory to our case and study moduli spaces of vector bundles over $\mathcal{X}$.

3.1 Coherent sheaves

Let $\mathcal{C}$ be a stacky curve, i.e. a regular, separated, finite type, connected Deligne-Mumford stack of dimension 1 over an algebraically closed field k of characteristic 0, such that it has a Zariski open subscheme.

In particular, $\mathcal{C}$ has only a finite number of stacky points. Denote by $\underline{p}$ the set of the stacky points of $\mathcal{C}$ and for $p \in \underline{p}$ denote by e_p the multiplicity of $\mathcal{C}$ at p. The stacky curve $\mathcal{C}$ admits a coarse moduli space C and is a root stack over the curve C. In particular, at each point $p \in \underline{p}$ the stack $\mathcal{C}$ has the same structure as observed in (1.2), namely

$$\begin{tikzcd} B\mu_{e_p} \arrow[r, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{C} \arrow[d, "\pi"] \\ \{p\} \arrow[r] & C. \end{tikzcd}$$

Let $\mathcal{F} \in \mathrm{Coh}(\mathcal{C})$ be a coherent sheaf on $\mathcal{C}$. As observed Section 2.3, its pullback to $B\mu_{e_p}$ is a μ_{e_p}-representation. Since μ_{e_p} is an abelian group, the representation decomposes as a direct sum of characters. In particular, denoting by k_i the character $\zeta \mapsto \zeta^i$, we get a decomposition

$$i^*\mathcal{F} = \bigoplus_{i \in \mathbb{Z}/e_p\mathbb{Z}} k_i^{m_{p,i}}.$$

Definition 3.1. [Taa24, Definition 1.2.7] With the notation as above, we can define the vector of *multiplicities* of $\mathcal{F}$ at p as

$$m_p(\mathcal{F}) \coloneqq (m_{p,0}, \dots m_{p,e_p-1}).$$

Denote by $\underline{m}(\mathcal{F})$ the collection of the multiplicity vectors at all the stacky points $p \in \underline{p}$.

The case we are interested in is $\mathcal{C} = \mathcal{X} = \sqrt[e]{p/X}$. As proven in Section 2.3, the tautological sheaf $\mathcal{O}(\frac{1}{e}p)$ corresponds to the tautological character $\zeta \mapsto \zeta$, therefore

$$m_p(\mathcal{O}(\tfrac{1}{e}p)) = (0, 1, 0, \dots 0).$$

Since pullback commutes with tensor products, and $k_i \otimes k_j = k_{i+j}$, we see that tensoring with the tautological sheaf acts as a shift operator on the multiplicity vector.

We now want to understand how the multiplicities interact wih the pullback and pushforward functors π^* and π_*.

Proposition 3.2. [Taa24, Example 1.2.9] If F is a coherent sheaf on C, then $m_p(\pi^*F) = (n, 0, \dots 0)$ for any $p \in \underline{p}$. Namely, the pullback π^*F is endowed with the trivial μ_{e_p} action at each p.

Proposition 3.3. [Taa24, Corollary 1.2.6] If F is a coherent sheaf on C, then the natural morphism $F \to \pi_*\pi^*F$ is an isomorphism.

Proposition 3.4. [Taa24, Theorem 1.2.10] If $\mathcal{F}$ is a coherent sheaf on $\mathcal{C}$ such that $m_p(\mathcal{F}) = (n, 0, \dots, 0)$ for all $p \in \underline{p}$, then the natural morphism $\pi^*\pi_*\mathcal{F} \to \mathcal{F}$ is an isomorphism.

Notice that if $\mathcal{E}$ is a vector bundle on $\mathcal{C}$, we have that

$$\sum_{i=0}^{e_p-1} m_{p,i}(\mathcal{E}) = \mathrm{rk}\,\mathcal{E} \tag{3.1}$$

for any $p \in \underline{p}$. In particular, if $\mathcal{L}$ is a line bundle, then for each $p \in \underline{p}$ there exists a unique index a_p such that $m_{p,a_p} = 1$, while $m_{p,i} = 0$ for all other indices $i \neq a_p$.

Theorem 3.5. [Cad07, Corollary 3.2.1; Taa24, Corollary 1.2.11] Let $\mathcal{L}$ be a line bundle on $\mathcal{C}$. For each stacky point $p \in \underline{p}$ let a_p be the unique number such that $m_{p,a_p} \neq 0$. Then there exist a unique (up to isomorphism) line bundle L on C such that

$$\mathcal{L} \cong \pi^* L \otimes \bigotimes_{p} \mathcal{O}(\tfrac{1}{e_p} p)^{\otimes a_p}. \tag{3.2}$$

In particular, L is characterized by the property $\pi_* \mathcal{L} \cong L$.

Remark 3.6. Using the decomposition (3.2) and Proposition 3.3, we get the property $\pi_* \mathcal{O}(\frac{k}{e_p} p) = \mathcal{O}(\lfloor \frac{k}{e_p} \rfloor p)$.

3.2 Degree of vector bundles on a stacky curve

Let us recall the definition of the Groethendieck group. If X is a Noetherian scheme, $K(X)$ is the abelian group generated by isomorphism classes of coherent sheaves on X, modulo the relations given by $\mathcal{F} = \mathcal{F}' + \mathcal{F}''$ if there is a short exact sequence

$$0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{F}'' \to 0.$$

Theorem 3.7. If C is a regular, separated, geometrically connected curve, then the pair $(\det, \mathrm{rk})\colon K(C) \to \mathrm{Pic}_C \oplus \mathbb{Z}$ defines an isomorphism.

Theorem 3.8. [Taa24, Theorem 1.2.27] Let $\mathcal{C}$ be a stacky curve, with stacky points $\underline{p}$. Then the following is a group isomorphism

$$K(\mathcal{C}) \xrightarrow{\left(\det \circ \pi_*,\ \mathrm{rk},\ (m_{p,i})_{i=1}^{e_p-1}\right)} \mathrm{Pic}(C) \oplus \mathbb{Z} \oplus \bigoplus_{p \in \underline{p}} \mathbb{Z}^{e_p - 1}.$$

Definition 3.9. [Taa24, Definition 1.2.28] This theorem allows us to define the determinant on $\mathcal{C}$ as

$$\begin{aligned} \det\colon K(\mathcal{C}) \xrightarrow{\left(\det \circ \pi_*,\ (m_{p,i})_{i=1}^{e_p-1}\right)} \mathrm{Pic}(C) \oplus \bigoplus_{p \in \underline{p}} \mathbb{Z}^{e_p-1} &\longrightarrow \mathrm{Pic}(\mathcal{C}) \\ (L, \underline{m}) \quad &\mapsto \ \pi^* L \otimes \mathcal{O}(\tfrac{i}{e_p} p)^{\otimes m_{p,i}}. \end{aligned}$$

Remark 3.10. Note that the above definition respects the property that a line bundle is its own determinant. Actually, the map det is the unique group homomorphism $K(\mathcal{C}) \to \mathrm{Pic}(\mathcal{C})$ such that $\det(\mathcal{L}) = \mathcal{L}$ for any line bundle $\mathcal{L}$ on $\mathcal{C}$. Indeed, isomorphism between $\det(\mathcal{L})$ and $\mathcal{L}$ follows from the decomposition (3.2).

Remark 3.11. Using the splitting principle and the above observation, we can prove that for vector bundles on $\mathcal{C}$ this is indeed the determinant map, i.e.

$$\det(\mathcal{E}) = \bigwedge_{i=0}^{\mathrm{rk}\,\mathcal{E}} \mathcal{E}.$$

Therefore, the usual formulas for determinant of vector bundles hold, for example

$$\det(\mathcal{E}_1 \otimes \mathcal{E}_2) = \det(\mathcal{E}_1)^{\operatorname{rk}\mathcal{E}_2} \det(\mathcal{E}_2)^{\operatorname{rk}\mathcal{E}_1}. \tag{3.3}$$

Remark 3.12. Suppose $\underline{p} = (p_1, \dots, p_n)$. Theorem 3.5 gives an isomorphism

$$\operatorname{Pic}(\mathcal{C}) \cong \operatorname{Pic}(C)\left[\frac{\mathcal{O}(p_1)}{e_1}, \dots, \frac{\mathcal{O}(p_n)}{e_n}\right].$$

We can use this isomorphism to define the degree of a coherent sheaf on $\mathcal{C}$.

Definition 3.13. [Taa24, Definition 1.2.29] Consider the composition

$$K(\mathcal{C}) \xrightarrow{\det} \operatorname{Pic}(\mathcal{C}) \xrightarrow{\cong} \operatorname{Pic}(C)\left[\frac{\mathcal{O}(p_1)}{e_1}, \dots, \frac{\mathcal{O}(p_n)}{e_n}\right] \xrightarrow{\deg} \mathbb{Z}\left[\frac{1}{e_1}, \dots, \frac{1}{e_n}\right] \subseteq \mathbb{Q}.$$

The degree of a coherent sheaf $\mathcal{F}$ is defined as its image under this composition.

Remark 3.14. The degree map $\deg\colon K(\mathcal{C}) \to \mathbb{Q}$ is characterized by the fact that the pullback $\pi^*\colon K(C) \to K(\mathcal{C})$ is degree preserving.

Remark 3.15. Using Equation (3.3), we get the classical formula for the degree of the tensor products of line bundles. Namely, if $\mathcal{E}_1$ and $\mathcal{E}_2$ are vector bundles on $\mathcal{C}$, we get

$$\deg(\mathcal{E}_1 \otimes \mathcal{E}_2) = \operatorname{rk}\mathcal{E}_2 \deg \mathcal{E}_1 + \operatorname{rk}\mathcal{E}_1 \deg \mathcal{E}_2.$$

In general, if we want to compute the degree of vector bundle on $\mathcal{C}$, we need first to consider its determinant bundle. It will have a decomposition as in (3.2). Then it suffices to use the properties

$$\deg \pi^* L = \deg L$$
$$\deg \mathcal{O}(\tfrac{1}{e_p} p) = \tfrac{1}{e_p}$$

to conclude.

Putting these properties together, we get the following.

Proposition 3.16. [Taa24, Theorem 1.2.30] If $\mathcal{E}$ is a vector bundle on $\mathcal{C}$, its degree can be computed via

$$\deg \mathcal{E} = \deg(\pi_* \mathcal{E}) + \sum_{p \in \underline{p}} \frac{1}{e_p} \sum_{i=0}^{e_p - 1} i\, m_{e_p, i}. \tag{3.4}$$

In particular, for line bundles we get $\deg \pi_* \mathcal{L} = \lfloor \deg \mathcal{L} \rfloor$.

3.3 Numerical invariants for vector bundles

In Sections 3.1 and 3.2 we introduced two numerical invariants associated to a vector bundle $\mathcal{E}$ on $\mathcal{C}$, the multiplicities vector $\underline{m}(\mathcal{E})$ and the degree $\deg \mathcal{E}$. They

are correlated by Equation (3.4). More explicitly, there is another numerical invariant we can associate to $\mathcal{E}$, namely

$$d_0(\mathcal{E}) \coloneqq \deg(\pi_*\mathcal{E}).$$

Then Equation (3.4) shows that the datum $(\underline{m}(\mathcal{E}), \deg \mathcal{E})$ is equivalent to the datum $(\underline{m}(\mathcal{E}), d_0(\mathcal{E}))$.

As shown in [Taa24, Section 1.3], these are the only the numerical invariants appearing in the coefficients of the Hilbert polynomial of $\mathcal{E}$, therefore they will determine a connected component of the moduli stack of vector bundles on $\mathcal{C}$.

For the sake of completeness, let us also introduce a third equivalent pair of numerical invariants. Of course, in the study of vector bundles another invariant to consider is the rank. As observed in Equation (3.1), for all $p \in \underline{p}$ we have

$$\operatorname{rk} \mathcal{E} = \sum_{i=0}^{e_p-1} m_{p,i}.$$

Let us also introduce another notion of degree, that captures the behavior of $\mathcal{E}$ at the stacky points.

Definition 3.17. [Taa24, Definition 1.2.7] Let $\mathcal{F}$ be a coherent sheaf on $\mathcal{C}$. The *twisted degrees* of $\mathcal{F}$ at $p \in \underline{p}$ are

$$d_{p,i}(\mathcal{F}) \coloneqq \deg\left(\pi_*\mathcal{F} \otimes \mathcal{O}(\tfrac{i}{e_p}p)\right).$$

Write $d_p(\mathcal{F}) = (d_{p,0}, \dots d_{p,e_p-1})$ for the vector of twisted degrees at p and $\underline{d}(\mathcal{F})$ for the collection of all twisted degrees.

Notice that by definition we have $d_{p,0}(\mathcal{F}) = d_0(\mathcal{F})$ for all $p \in \underline{p}$.

Finally, for a vector bundle $\mathcal{E}$, the datum $(\underline{m}(\mathcal{E}),\ d_0(\mathcal{E}))$ is equivalent to the datum $(\operatorname{rk}\mathcal{E},\ \underline{d}(\mathcal{E}))$. Indeed we have the following result.

Proposition 3.18. [Taa24, Corollary 1.2.21] Let $\mathcal{E}$ be a vector bundle on $\mathcal{C}$ and $p \in \underline{p}$ a stacky point. Then $m_{p,i} = d_{p,i} - d_{p,i-1}$ for $i = 1, \dots e_p - 1$ and $m_{p,0} = \operatorname{rk}\mathcal{E} - \sum_{i=1}^{e_p-1} m_{p,i}$.

In conclusion, to a vector bundle $\mathcal{E}$ we can associate three equivalent pairs of numerical invariants, i.e. the pairs $(\underline{m}(\mathcal{E}), \deg \mathcal{E})$, $(\underline{m}(\mathcal{E}), d_0(\mathcal{E}))$ and $(\operatorname{rk}\mathcal{E}, \underline{d}(\mathcal{E}))$. Each of these pairs determines the coefficients of the Hilbert polynomial of $\mathcal{E}$ and therefore determines the connected component of $\mathcal{E}$ in the moduli space of vector bundles on $\mathcal{C}$.

3.4 Cotangent sheaf

Definition 3.19. Let $f\colon \mathcal{X} \to \mathcal{Y}$ be a morphism of Deligne Mumford stacks. The cotangent sheaf $\Omega_{\mathcal{X}/\mathcal{Y}}$ is the sheaf on the étale site of $\mathcal{X}$ defined as follows. Let $\mathcal{I}$ be the kernel of the morphism $\mathcal{O}_{\mathcal{X}} \otimes_{f^{-1}\mathcal{O}_{\mathcal{Y}}} \mathcal{O}_{\mathcal{X}} \to \mathcal{O}_{\mathcal{X}}$. Then $\Omega_{\mathcal{X}/\mathcal{Y}} \coloneqq \mathcal{I}/\mathcal{I}^2$.

Remark 3.20. This has the usual properties of the cotangent sheaf. In particular, we have the following.

- If we have morphisms $\mathcal{X} \xrightarrow{f} \mathcal{Y} \xrightarrow{g} \mathcal{Z}$, then we get an exact sequence of sheaves on $\mathcal{X}$
$$f^*\Omega_{\mathcal{Y}/\mathcal{Z}} \to \Omega_{\mathcal{X}/\mathcal{Z}} \to \Omega_{\mathcal{X}/\mathcal{Y}} \to 0.$$
Moreover, if $\mathcal{O}_{\mathcal{X}}$ is locally free over $f^{-1}\mathcal{O}_{\mathcal{Y}}$, then the sequence can be extended with 0 to the left.

- If $i\colon \mathcal{Y} \to \mathcal{X}$ is a closed immersion with ideal sheaf $\mathcal{J}$, then the sequence
$$\mathcal{J}/\mathcal{J}^2 \to i^*\Omega_{\mathcal{X}} \to \Omega_{\mathcal{Y}} \to 0$$
is exact.

Theorem 3.21. [Taa24, Theorem 1.2.34] Let $\pi\colon \mathcal{C} \to C$ be a smooth stacky curve, with set of stacky points $\underline{p}$. Then
$$\Omega_{\mathcal{C}} \cong \pi^*\Omega_C \otimes \bigotimes_{p\in\underline{p}} \mathcal{O}(\tfrac{1}{e_p}p)^{\otimes e_p - 1}.$$

4 Moduli spaces of vector bundles

We now come to the study of $\mathcal{B}un_n(\mathcal{C})$, the moduli stack of rank n vector bundles on $\mathcal{C}$. As mentioned in Section 3.3, we have three equivalent pairs of invariants associated to a vector bundle on $\mathcal{C}$. Each of these pairs determines a connected component of the moduli stack. Indeed, we have the following.

Theorem 4.1. [Taa24, Corollary 2.0.9, Theorem 2.0.10] Let $\mathcal{B}un_n^{\underline{d}}(\mathcal{C})$ be the substack of $\mathcal{B}un(\mathcal{C})$ defined by
$$\mathcal{B}un_n^{\underline{d}}(\mathcal{C}) := \langle \mathcal{E} \in \mathcal{B}un(\mathcal{C}) \mid (\operatorname{rk}\mathcal{E}, \underline{d}(\mathcal{E})) = (n, \underline{d})\rangle .$$
Then the substacks $\mathcal{B}un_n^{\underline{d}}(\mathcal{C})$ are the connected components of $\mathcal{B}un(\mathcal{C})$. Moreover, the stack $\mathcal{B}un(\mathcal{C})$ is smooth, so the connected components coincide with the irreducible components and they are smooth.

Let $\mathcal{B}un_{\underline{m}}^d$ be the substack of $\mathcal{B}un(\mathcal{C})$ defined by
$$\mathcal{B}un_{\underline{m}}^d(\mathcal{C}) := \langle \mathcal{E} \in \mathcal{B}un(\mathcal{C}) \mid (\underline{m}(\mathcal{E}), \deg\mathcal{E}) = (\underline{m}, d)\rangle .$$
Using Theorem 4.1, if we fix the rank n, we get a decomposition
$$\mathcal{B}un_n(\mathcal{C}) = \coprod_{\underline{d}} \mathcal{B}un_n^{\underline{d}}(\mathcal{C}) = \coprod_{(\underline{m},d)} \mathcal{B}un_{\underline{m}}^d(\mathcal{C}),$$
where the latter union ranges over the pairs $(\underline{m}, d)$ such that for all $p \in \underline{p}$ we have $\sum_{i=0}^{e_p-1} m_{p,i} = n$.

Consider now the case where $C = X$ is a smooth projective genus g curve and $\mathcal{C} = \mathcal{X} = \sqrt[n]{p/X}$ has a unique stacky point of multiplicity $e = n$. Then, for a vector bundle $\mathcal{E}$ on $\mathcal{X}$ the multiplicity $\underline{m}$ is just the vector $(m_{p,0}, \dots m_{p,n-1})$ of multiplicities of $\mathcal{E}$ at p. Equivalently, it classifies the representation obtained as $\mathcal{E}|_{B\mu_n}$.

We now want to show the following result.

Theorem 4.2. There is an equivalence of categories

$$\mathcal{B}un_n^d(X) \cong \mathcal{B}un^0_{(0,\dots,0,n,0,\dots,0)}(\mathcal{X}) \tag{4.1}$$

preserving (semi)stability, where on the right hand side the n is in the position $-d(\operatorname{mod} n)$.

In order to simplify the notation, let us denote the multiplicity vector on the right hand side by n_{-d}. Similarly, we will denote $(n, 0, \dots, 0)$ by n_0.

The most relevant step in the proof is the following

Lemma 4.3. The pullback and pushforward functors define equivalences

$$\mathcal{B}un_n^d(X) \cong \mathcal{B}un_{n_0}^d(\mathcal{X}),$$

which preserve (semi)stability.

Proof. First of all, we need to observe that the pullback and pushforward functors map vector bundles to vector bundles, preserving the invariants.

For π^*, it is known that the pullback maps vector bundles to vector bundles of the same rank and the degree on $\mathcal{X}$ was define so that π^* would preserve the degree, see Remark 3.14. Moreover, we noticed in Proposition 3.2 that, for $F \in \mathcal{B}un_n(X)$, the multiplicity of π^*F at p is $(n, 0, \dots, 0)$, or equivalently that $\pi^*F|_{B\mu_n}$ is the trivial representation. Therefore π^* restricts as a well defined functor

$$\pi^* \colon \mathcal{B}un_n^d(X) \to \mathcal{B}un_{n_0}^d(\mathcal{X}).$$

In the other direction, we first need to show that the pushforward $\pi_* \colon \operatorname{QCoh}(\mathcal{X}) \to \operatorname{QCoh}(X)$ maps vector bundles to vector bundles. This is [AV02, Theorem 2.3.4]. Since the rank of a vector bundle on a connected stack is the same at each point and π is an isomorphism away from p, it is also clear that π_* maps rank n vector bundles to rank n vector bundles. Finally, Equation (3.4) shows that, if the multiplicities vector is $(n, 0, \dots, 0)$, the degree of $\pi_*\mathcal{E}$ equals the degree of $\mathcal{E}$. Therefore, π_* restricts to a well defined functor

$$\pi_* \colon \mathcal{B}un_{n_0}^d(\mathcal{X}) \to \mathcal{B}un_n^d(X).$$

The fact that these are inverse equivalences are just Propositions 3.3 and 3.4.

Notice that these are equivalences of categories of vector bundles, preserving both the rank and the degree. Therefore, they preserve (semi)stability. □

Now the proof of Theorem 4.2 follows easily.

Proof. Notice now that tensoring with $\mathcal{O}(-\frac{1}{n}p)^{\otimes d}$ defines an equivalence $\mathcal{B}un^d_{n_0}(\mathcal{X}) \cong \mathcal{B}un^0_{n_{-d}}(\mathcal{X})$. Explicitly,

$$\begin{tikzcd} \mathcal{B}un^d_{n_0}(\mathcal{X}) \arrow[r, bend left, "-\otimes\mathcal{O}(-\frac{d}{n}p)"] \arrow[r, phantom, "\cong"] & \mathcal{B}un^0_{n_{-d}}(\mathcal{X}). \arrow[l, bend left, "\mathcal{O}(\frac{d}{n}p)\otimes-"] \end{tikzcd}$$

Clearly the two maps are inverse to each other. The only nontrivial thing to show is that they respect the defined invariants. Firstly, it was already observed in Section 3.1 that the tautological line bundle acts as a shift operator on the multiplicities. Therefore, n_0 becomes n_{-d} after tensoring and vice versa. Secondly, using Remark 3.15, we get

$$\deg\left(\mathcal{E}\otimes\mathcal{O}(-\tfrac{d}{n}p)\right) = \deg\mathcal{E} + n\left(-\frac{d}{n}\right) = 0.$$

Finally, notice how these equivalences also preserve (semi)stability. Indeed, let $\mathcal{E} \in \mathcal{B}un^d_{n_0}(\mathcal{X})$ be a semistable vector bundle. Since the indicated functors are equivalences, in particular they preserve subbundles. Therefore, it suffices to show that if $\mathcal{E}' \subseteq \mathcal{E}$ is a subbundle such that $\mu(\mathcal{E}') \leq \mu(\mathcal{E})$, then $\mu(\mathcal{E}' \otimes \mathcal{O}(-\frac{d}{n}p)) \leq \mu(\mathcal{E}\otimes\mathcal{O}(-\frac{d}{n}p))$. This follows from the simple computation

$$\mu(\mathcal{E}'\otimes\mathcal{O}(-\tfrac{d}{n}p)) = \frac{\deg\mathcal{E}' - \operatorname{rk}\mathcal{E}'\frac{d}{n}}{\operatorname{rk}\mathcal{E}'} = \mu(\mathcal{E}') - \frac{d}{n}.$$

Repeating the same computation for $\mathcal{E}$, one shows that $\mu(\mathcal{E}'\otimes\mathcal{O}(-\frac{d}{n}p)) \leq \mu(\mathcal{E}\otimes\mathcal{O}(-\frac{d}{n}p))$. The computation in the other direction is completely analogous. $\square$

Putting everything together, we get an equivalence of categories

$$\begin{tikzcd} \mathcal{B}un^d_n(X) \arrow[r, shift left, "\pi^*(-)\otimes\mathcal{O}(-\frac{d}{n}p)"] & \mathcal{B}un^0_{n_{-d}}(\mathcal{X}), \arrow[l, shift left, "\pi_*(\mathcal{O}(-\frac{d}{n}p)\otimes-)"] \end{tikzcd}$$

that restricts to an equivalence

$$\begin{tikzcd} \mathcal{B}un^{d,ss}_n(X) \arrow[r, shift left, "\pi^*(-)\otimes\mathcal{O}(-\frac{d}{n}p)"] & \mathcal{B}un^{0,ss}_{n_{-d}}(\mathcal{X}). \arrow[l, shift left, "\pi_*(\mathcal{O}(-\frac{d}{n}p)\otimes-)"] \end{tikzcd}$$

5 Dolbeault moduli spaces

Recall tha the Dolbeault moduli space is the moduli space of semistable Higgs bundles, i.e.

$$\operatorname{Higgs}^d_n(X) \coloneqq \left\langle (\mathcal{E},\varphi) \;\middle|\; \mathcal{E}\in\mathcal{B}un^d_n(X),\ \varphi\colon \mathcal{E}\to\mathcal{E}\otimes\Omega_X \right\rangle,$$
$$\mathcal{M}^d_{n,Dol}(X) \coloneqq \operatorname{Higgs}^{d,ss}_n(X).$$

We can define similarly $\mathrm{Higgs}^d_{\underline{m}}(\mathcal{X})$ and $\mathcal{M}^d_{\underline{m},Dol}(\mathcal{X})$.

We now want to extend Theorem 4.2 to the Dolbeault moduli spaces and prove the following.

Theorem 5.1. There is an equivalence of categories

$$\mathcal{M}^d_{n,Dol}(X) \cong \mathcal{M}^0_{n_{-d},Dol}(\mathcal{X}).$$

As before, we will first extend **??** 4.3 to the moduli spaces of Higgs bundles and then Theorem 5.1 will follow. For convenience, we split the proof of Theorem 5.1 in intermediate lemmas.

Lemma 5.2. Under the isomorphism

$$\Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\tfrac{1}{n}p)^{\otimes n-1}$$

shown in Theorem 3.21, the canonical morphism

$$\pi^*\Omega_X \to \Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\tfrac{n-1}{n}p)$$

has the form $n\,\mathrm{id}_{\pi^*\Omega_X} \otimes s^{n-1}$, where s is the tautological section of $\mathcal{O}(\frac{1}{n}p)$.

Proof. The coarse moduli map $\pi\colon \mathcal{X} \to X$ locally around p has the following form

$$\begin{tikzcd}
\mathbb{A}^1 = \operatorname{Spec} k[x] \arrow[d, "q'"] \arrow[ddr, bend left, "q=\pi\circ q'"', "x\mapsto t=x^n"] & \\
\mathcal{X} = [\mathbb{A}^1/\mu_n] \arrow[d, "\pi"] & \\
X = \mathbb{A}^1/\mu_n \quad = & \operatorname{Spec} k[x^n] = \operatorname{Spec} k[t].
\end{tikzcd}$$

Notice that we have the relation

$$\mathrm{d}t = \mathrm{d}x^n = n\,x^{n-1}\ \mathrm{d}x,$$

therefore the canonical morphism $q^*\Omega_X \to \Omega_{\mathbb{A}^1}$ corresponds to

$$\begin{aligned} k[x] \otimes_{k[t]} k[t]\,\mathrm{d}t \ &\to\ k[x]\,\mathrm{d}x \\ f(x) \otimes \mathrm{d}t \quad &\mapsto\ f(x)\,n\,x^{n-1}\ \mathrm{d}x. \end{aligned}$$

Since q' is étale, it induces isomorphisms on the cotangent sheaves. Moreover, the section x corresponds via q' to the tautological section s of $\mathcal{O}(\frac{1}{n}p)$. In conclusion, we get that the canonical morphism $\pi^*\Omega_X \to \Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\frac{n-1}{n}p)$ has the indicated form. □

Now we can use this morphism to extend the pullback functor π^* to Higgs bundles. In particular, we need to lift the Higgs field. Define

$$\begin{aligned} (\pi^*, \pi^\#)\colon\ \mathrm{Higgs}^d_n(X) &\to \mathrm{Higgs}^d_{n_0}(\mathcal{X}) \\ (E,\varphi) &\mapsto (\pi^*E, \pi^\#\varphi), \end{aligned}$$

where $\pi^{\#}\varphi$ is defined as the composition

$$\pi^{\#}\varphi\colon \pi^*E \xrightarrow{\pi^*\varphi} \pi^*(E\otimes\Omega_X) = \pi^*E\otimes\pi^*\Omega_X \xrightarrow{\mathrm{id}\otimes n\ \mathrm{id}\otimes s^{n-1}} \pi^*E\otimes\pi^*\Omega_X\otimes\mathcal{O}(\tfrac{n-1}{n}p) = \pi^*E\otimes\Omega_{\mathcal{X}}.$$

Similarly, in the other direction define

$$\begin{aligned}(\pi_*,\pi_{\#})\colon\ \mathrm{Higgs}^d_{n_0}(\mathcal{X}) &\to \mathrm{Higgs}^d_n(X)\\ (\mathcal{E},\eta) &\mapsto (\pi_*\mathcal{E},\pi_{\#}\eta)\end{aligned}$$

as follows. From the equivalence (4.1), it follows that $\mathcal{E}=\pi^*E$, where $E\cong\pi_*\mathcal{E}$. By definition, η is a morphism in $\mathrm{Hom}_{\mathcal{X}}(\mathcal{E},\mathcal{E}\otimes\Omega_{\mathcal{X}})$. Now, using Theorem 3.21, we find a chain of natural isomorphisms

$$\begin{aligned}\mathrm{Hom}_{\mathcal{X}}(\mathcal{E},\,\mathcal{E}\otimes\Omega_{\mathcal{X}}) &\cong \mathrm{Hom}_{\mathcal{X}}\left(\pi^*E,\,\pi^*E\otimes\pi^*\Omega_X\otimes\mathcal{O}(\tfrac{n-1}{n}p)\right)\\ &\cong \mathrm{Hom}_X\left(E,\,\pi_*(\pi^*(E\otimes\Omega_{\mathcal{X}})\otimes\mathcal{O}(\tfrac{n-1}{n}p))\right) &(5.1)\\ &\cong \mathrm{Hom}_X\left(E,\,E\otimes\Omega_X\otimes\pi_*\mathcal{O}(\tfrac{n-1}{n}p)\right) &(5.2)\\ &\cong \mathrm{Hom}_X\left(E,\,E\otimes\Omega_X\right), &(5.3)\end{aligned}$$

where (5.1) follows by the adjunction $\pi^*\dashv\pi_*$, (5.2) by the projection formula and (5.3) by the isomorphism $\pi_*\mathcal{O}(\frac{n-1}{n}p)\cong\mathcal{O}_X$.
Composing the isomorphisms, we get

$$\begin{aligned}\mathrm{Hom}_X(E,E\otimes\Omega_X) &\xrightarrow{\cong} \mathrm{Hom}_{\mathcal{X}}(\mathcal{E},\mathcal{E}\otimes\Omega_{\mathcal{X}})\\ \psi &\mapsto n\,\pi^*\psi\otimes s^{n-1}.\end{aligned}$$

This shows that there exists a unique $\psi\in\mathrm{Hom}_X(E,E\otimes\Omega_X)$ such that $\eta = n\,\pi^*\psi\otimes s^{n-1}$, where in particular we have $\psi=\frac{1}{n}\,\pi_*\eta$. Therefore, we can define

$$\pi_{\#}\eta := \frac{1}{n}\,\pi_*\eta.$$

This gives a well defined functor $(\pi_*,\pi_{\#})$.

We can now phrase the analogue of **??** 4.3.

Lemma 5.3. The functors $(\pi^*,\pi^{\#})$ and $(\pi_*,\pi_{\#})$ define inverse equivalences of categories

$$\begin{tikzcd}\mathrm{Higgs}^d_n(X) \arrow[r, bend left, "(\pi^*,\pi^{\#})"] \arrow[r, phantom, "\cong"] & \mathrm{Higgs}^d_{n_0}(\mathcal{X}). \arrow[l, bend left, "(\pi_*,\pi_{\#})"]\end{tikzcd}$$

Proof. At the level of vector bundles this is **??** 4.3. We need to show that the same holds for the Higgs fields. Indeed, computing the two compositions we get

$$\varphi \xmapsto{\pi^{\#}} n\,\pi^*\varphi\otimes s^{n-1} \xmapsto{\pi_{\#}} \frac{1}{n}\,\pi_*(n\,\pi^*\varphi\otimes s^{n-1}) = \varphi$$

$$\eta \xmapsto{\pi_{\#}} \frac{1}{n}\,\pi_*\eta \xmapsto{\pi^{\#}} n\,\pi^*(\frac{1}{n}\,\pi_*\eta)\otimes s^{n-1} = n\,\pi^*\psi\otimes s^{n-1} = \eta.$$

□

As observed earlier, tensoring with a line bundle induces an equivalence, so we get

$$\mathrm{Higgs}_n^d(X) \underset{(\pi_*,\pi^\#)(\mathcal{O}(-\frac{d}{n}p)\otimes -)}{\overset{(\pi^*,\pi^\#)(-)\otimes\mathcal{O}(-\frac{d}{n}p)}{\rightleftarrows}} \mathrm{Higgs}_{n-d}^0(\mathcal{X}).$$

Finally, we noticed that (semi)stability is preserved, therefore we get the desired equivalence

$$\mathcal{M}_{n,Dol}^d(X) \underset{(\pi_*,\pi^\#)(\mathcal{O}(-\frac{d}{n}p)\otimes -)}{\overset{(\pi^*,\pi^\#)(-)\otimes\mathcal{O}(-\frac{d}{n}p)}{\rightleftarrows}} \mathcal{M}_{n-d,Dol}^0(\mathcal{X}). \tag{5.4}$$

This concludes the proof of Theorem 5.1.

6 Betti and De Rham moduli spaces

We now come to the study of the Betti and de Rham moduli spaces of the stacky curve $\mathcal{X}$. Recall that

$$\mathcal{M}_{n,DR}(\mathcal{X}) \coloneqq \langle (\mathcal{E},\nabla) \mid \mathcal{E}\in \mathcal{B}un_n(\mathcal{X}),\ \nabla\colon \mathcal{E}\to\mathcal{E}\otimes\Omega_X \text{ is a flat connection}\rangle$$
$$\mathcal{M}_{n,B}(\mathcal{X}) \coloneqq \mathrm{Rep}(\pi_1(\mathcal{X},x), GL_n).$$

The Riemann-Hilbert correspondence establishes an isomorphism of the coarse moduli spaces

$$M_{n,DR}(\mathcal{X})^{\mathrm{an}} \cong M_{n,B}(\mathcal{X})^{\mathrm{an}}. \tag{6.1}$$

If we fix a rank n and use the presentation of $\pi_1(\mathcal{X},x)$ found in (1.5), we get

$$M_{n,B}(\mathcal{X}) \cong \left\{ ((A_i,B_i)_{i=1}^g, C) \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i,B_i] = C,\ C^n = Id \right\} //GL_n. \tag{6.2}$$

As already observed for the moduli space of vector bundles $\mathcal{B}un(\mathcal{X})$ and for the Dolbeault moduli space, also the de Rham moduli space admits a decomposition in connected components

$$\mathcal{M}_{n,DR}(\mathcal{X}) = \coprod_{\underline{d}} \mathcal{M}_{n,DR}^{\underline{d}}(\mathcal{X}) = \coprod_{(\underline{m},d)} \mathcal{M}_{\underline{m},DR}^d(\mathcal{X}),$$

with the latter union ranging over the pairs $(\underline{m},d)$ such that $\sum_{i=0}^{n-1} m_i = n$. Recall that for a vector bundle to admit a flat connection it is a necessary condition that $\deg\mathcal{E}=0$. Therefore we can set $\mathcal{M}_{\underline{m},DR} \coloneqq \mathcal{M}_{\underline{m},DR}^0(\mathcal{X})$ and get

$$\mathcal{M}_{n,DR}(\mathcal{X}) = \coprod_{\underline{m}} \mathcal{M}_{\underline{m},DR}(\mathcal{X}).$$

This decomposition gives decomposition in connected components of the coarse moduli space

$$M_{n,DR}(\mathcal{X}) = \coprod_{\underline{m}} M_{\underline{m},DR}(\mathcal{X}). \tag{6.3}$$

We now want to understand how this decomposition of the coarse moduli space $M_{n,DR}$ translates to the Betti moduli space, and in particular to the character variety on the right hand side of (6.2).

6.1 The monodromy representation

The isomorphism in Equation (6.1) is obtained by mapping a pair $(\mathcal{E}, \nabla)$ to the associated monodromy representation at x. In particular, the matrices A_i, B_i, C in (6.2) are obtained by the action of the generators a_i, b_i, c of the fundamental group of $\mathcal{X}$, as expressed in Section 1.4, and are well defined up to conjugation.

The multiplicity vector $\underline{m}(\mathcal{E})$ depends only on the behavior of $\mathcal{E}$ at the stacky point, and therefore can be computed in a neighborhood of p isomorphic to $[\mathbb{A}^1/\mu_n]$. In particular, looking back at the Seifert-Van Kampen computation of $\pi_1(\mathcal{X}, x)$ in Section 1.4, we see that the decomposition (6.3) will translate on the character variety only in terms of conditions on the matrix C.

In particular, after restricting to $[\mathbb{A}^1/\mu_n]$, we can find the matrix C as obtained by the monodromy action of the generator of $\pi_1([\mathbb{A}^1/\mu_n], x) \cong \mu_n$ on $(\mathcal{E}|_{[\mathbb{A}^1/\mu_n]}, \nabla|_{[\mathbb{A}^1/\mu_n]})$.

Therefore, let us consider the case of a vector bundle $\mathcal{E}$ on $[\mathbb{A}^1/\mu_n]$, with a flat connection ∇. As mentioned in Section 1.4, the projection map

$$p\colon \mathbb{A}^1 \to [\mathbb{A}^1/\mu_n]$$

is a principal μ_n-bundle and it is the universal cover of $[\mathbb{A}^1/\mu_n]$. The datum of the pair $(\mathcal{E}, \nabla)$ is equivalent to the μ_n-equivariant pair $(p^*\mathcal{E}, p^*\nabla)$ on $\mathbb{A}^1$. The fundamental group of $[\mathbb{A}^1/\mu_n]$ is identified with μ_n and under this identification the monodromy action on $(\mathcal{E}, \nabla)$ is the pushforward of the μ_n-action on $(p^*\mathcal{E}, p^*\nabla)$.

Since $\mathbb{A}^1$ is simply connected and $p^*\nabla$ is a flat connection on $p^*\mathcal{E}$, the pair $(p^*\mathcal{E}, p^*\nabla)$ is actually equivalent to the trivial bundle, with the standard derivation. Moreover, since $\mathbb{A}^1$ is affine, this is just the sheafification of the pair $(\mathbb{C}[x]^n, \mathrm{d})$. The μ_n-equivariant structure corresponds to an action

$$\mu_n \to \mathrm{Aut}_{\mathbb{C}[x]}((\mathbb{C}[x]^n, \mathrm{d})).$$

Notice that an automorphism $\varphi\colon \mathbb{C}[x]^n \to \mathbb{C}[x]^n$ fixes the differential d if and only if $\mathrm{d}(\varphi) = 0$. Therefore we have the following isomorphisms

$$\begin{array}{ccc} \mathrm{Aut}_{\mathbb{C}[x]}(\mathbb{C}[x]^n) & \cong & GL_n(\mathbb{C}[x]) \\ \cup\!\mid & & \cup\!\mid \\ \mathrm{Aut}_{\mathbb{C}[x]}((\mathbb{C}[x]^n, \mathrm{d})) & \cong & GL_n(\mathbb{C}). \end{array}$$

In particular, the μ_n-equivariant structure on $(p^*\mathcal{E}, p^*\nabla)$ gives an action

$$\rho\colon \mu_n \to GL_n(\mathbb{C}),$$

i.e. an n-dimensional representation of μ_n, which is exactly the monodromy representation of the pair $(\mathcal{E}, \nabla)$.

We now want to compare this with the representation obtained as $\mathcal{E}|_{B\mu_n}$. Consider

$$\begin{tikzcd} * = \{0\} \arrow[r, hook, "i"] \arrow[d] & \mathbb{A}^1 \arrow[d, "p"] \\ B\mu_n \arrow[r, hook, "j"'] & {[\mathbb{A}^1/\mu_n]} \end{tikzcd}$$

The representation associated to $\mathcal{E}|_{B\mu_n}$ is obtained by the μ_n-equivariant structure of $(p|_{\{0\}})^*\mathcal{E}|_{B\mu_n}$, corresponding to a morphism

$$\mu_n \to \mathrm{Aut}_{\{0\}}((p|_{\{0\}})^*\mathcal{E}|_{B\mu_n}) = \mathrm{Aut}_{\{0\}}(i^*p^*\mathcal{E}).$$

Under the identifications $\mathrm{QCoh}(\mathbb{A}^1) \cong \mathbb{C}[x]\text{-}\,\mathrm{Mod}$ and $\mathrm{QCoh}(*) \cong \mathbb{C}\text{-}\,\mathrm{Mod}$, the restriction i^* is just the morphism of evaluation at 0. Since $p^*\mathcal{E}$ is the trivial rank n vector bundle, we get

$$\begin{tikzcd} \mu_n \arrow[r, "\rho'"] \arrow[dr, "\sigma"'] & GL_n(\mathbb{C}[x]) \arrow[d, "\mathrm{ev}_0"] & \cong & \mathrm{Aut}_{\mathbb{A}^1}(p^*\mathcal{E}) \arrow[d, "i^*"] \\ & GL_n(\mathbb{C}) & \cong & \mathrm{Aut}_{\{0\}}(i^*p^*\mathcal{E}). \end{tikzcd}$$

But ρ' must factor through the inclusion

$$\mathrm{Aut}_{\mathbb{A}^1}((p^*\mathcal{E}, \nabla)) \subseteq \mathrm{Aut}_{\mathbb{A}^1}(p^*\mathcal{E}),$$

so we get

$$\begin{tikzcd} \mu_n \arrow[r, "\rho"] \arrow[rr, bend left, "\rho'"] \arrow[drr, "\sigma"'] & GL_n(\mathbb{C}) & \subseteq & GL_n(\mathbb{C}[x]) \arrow[d, "\mathrm{ev}_0"] \\ & & & GL_n(\mathbb{C}). \end{tikzcd}$$

Clearly the composition $GL_n(\mathbb{C}) \subseteq GL_n(\mathbb{C}[x]) \xrightarrow{\mathrm{ev}_0} GL_n(\mathbb{C})$ is the identity, so the representations ρ and σ coincide.

In conclusion, we obtained the following result.

Proposition 6.1. The monodromy representation of a pair $(\mathcal{E}, \nabla)$ coincides with the representation obtained as $\mathcal{E}|_{B\mu_n}$.

6.2 Betti moduli space

Applying Proposition 6.1 to the computation of the coarse moduli spaces, we get that

$$M_{\underline{m},B}(\mathcal{X}) \cong \left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = C_{\underline{m}} \right\} //GL_n,$$

where $C_{\underline{m}}$ is a matrix representing the action of the generator of μ_n in the representation defined by the multiplicity vector $\underline{m}$. In particular, $C_{\underline{m}}$ can be taken as a diagonal matrix with exactly m_l entries equal to $\mathrm{e}^{\frac{2\pi \mathrm{i}}{n} l}$.

In the case $\underline{m} = n_{-d}$, we get

Proposition 6.2. The n_{-d} stratum of coarse moduli space of the Betti moduli space of $\mathcal{X}$ is isomorphic to the following variety

$$M_{n_{-d},B}(\mathcal{X}) \cong \left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} //GL_n.$$

7 The nonabelian Hodge homeomorphism

In [Sim11, §8, 9] Simpson showed, using a simplicial resolution, that the moduli spaces of vector bundles over a smooth Deligne-Mumford stack respect the Riemann-Hilbert correspondence and the non-abelian Hodge homeomorphism.

More specifically, for a smooth Deligne-Mumford stack $\mathcal{Y}$ he established the Riemann-Hilbert isomorphism

$$M_{n,DR}(\mathcal{Y})^{\mathrm{an}} \cong M_{n,B}(\mathcal{Y})^{\mathrm{an}}, \tag{7.1}$$

and the non-abelian Hodge homeomorphism

$$M_{n,Dol}(\mathcal{Y})^{\mathrm{top}} \cong M_{n,B}(\mathcal{Y})^{\mathrm{top}}. \tag{7.2}$$

Let $\mathcal{Y} = \mathcal{X}$ be the stacky curve that we have studied so far. Of course, (7.1) and (7.2) respect the decomposition in connected components, so they restrict to

$$M_{\underline{m},DR}(\mathcal{X})^{\mathrm{an}} \cong M_{\underline{m},B}(\mathcal{Y})^{\mathrm{an}},$$
$$M^0_{\underline{m},Dol}(\mathcal{X})^{\mathrm{top}} \cong M_{\underline{m},B}(\mathcal{X})^{\mathrm{top}}.$$

Using the observations made in Section 6, these translate to homeomorphisms

$$M^0_{\underline{m},Dol}(\mathcal{X})^{\mathrm{top}} \cong \left(\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = C_{\underline{m}} \right\} //GL_n \right)^{\mathrm{top}},$$

and in particular to

$$M^0_{n_{-d},Dol}(\mathcal{X})^{\mathrm{top}} \cong \left(\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} // GL_n \right)^{\mathrm{top}}.$$

On the other hand, the equivalence (5.4) proven in Section 5 induces an isomorphism of the coarse moduli spaces

$$M^d_{n,Dol}(X) \cong M^0_{n_{-d},Dol}(\mathcal{X}).$$

Proposition 7.1. Putting the above homeomorphisms together, we get an homeomorphism

$$M^d_{n,Dol}(X)^{\mathrm{top}} \cong \left(\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} // GL_n \right)^{\mathrm{top}},$$

therefore the space on the right hand side can be interpreted as version of the character variety for bundles of nonzero degree on X.

The picture can be explained with more clarity by embedding both $M^0_{n,Dol}$ and $M^0_{n,DR}$ in a bigger moduli space $M^0_{n,Hdg}$. Indeed, for a fixed complex number λ, define the notion of λ-connection on a vector bundle $\mathcal{E}$ as a morphism $\nabla \colon \mathcal{E} \to \mathcal{E} \otimes \Omega$, respecting the Leibniz rule up to the scalar λ, namely

$$\nabla(f \cdot s) = s \otimes \mathrm{d}f + \lambda f \cdot \nabla(s).$$

Then $M^0_{n,Hdg}(\mathcal{X})$ is the moduli space parametrising isomorphism classes of triples $(\mathcal{E}, \nabla, \lambda)$. Notice that

- the multplicative group $\mathbb{G}_m$ acts on $M^0_{n,Hdg}$ by
$$t \cdot (\mathcal{E}, \nabla, \lambda) = (\mathcal{E}, t\nabla, t\lambda);$$
- there is a $\mathbb{G}_m$-equivariant projection
$$\begin{aligned} \pi \colon M^0_{n,Hdg} &\to \mathbb{A}^1 \\ (\mathcal{E}, \nabla, \lambda) &\mapsto \lambda, \end{aligned}$$
where $\mathbb{G}_m$ acts on $\mathbb{A}^1$ with weight one;
- the fibers $\pi^{-1}(\lambda)$ are all isomorphic for $\lambda \neq 0$ to $\pi^{-1}(1) = M^d_{n,DR}$, while $\pi^{-1}(0) = M^d_{n,Dol}$.

This picture restricts to any connected component, determined by numerical invariants. Summing up, letting $M_{Hdg} \coloneqq M^0_{n_{-d},Hdg}(\mathcal{X}), M_{Dol} \coloneqq M^0_{n_{-d},Dol}(\mathcal{X})$ and $M_{Dr} \coloneqq M^0_{n_{-d},DR}(\mathcal{X})$, we have the following diagram

$$\begin{tikzcd} M_{Dol} \arrow[r, hook, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & M_{Hdg} \arrow[d, "\pi"] & M_{DR} \arrow[l, hook', "j"'] \arrow[r, "\psi", "\cong"'] \arrow[d] \arrow[dl, phantom, "\llcorner"] & M_B \\ \{0\} \arrow[r] & \mathbb{A}^1 & \{1\}, \arrow[l] & \end{tikzcd}$$

where the inclusions i and j are algebraic, while ψ is an diffeomorphism of complex manifolds that is only defined analytically via the Riemann-Hilbert correspondence (7.1).

We want to compare the singular cohomology with rational coefficients of M_{Dol}, M_{Hdg}, M_{DR} and their Hodge structures. The idea to use the Białynicki-Birula decomposition was inspired by Nakajima's appendix to [CV04].

Notice that M_{Hdg} is a *semiprojective variety*, according to the terminology used in [HR15]. This means that it is a quasi-projective variety with a $\mathbb{G}_m$-action such that

- the fixed point locus $M_{Hdg}^{\mathbb{G}_m}$ is proper;
- the limit point $\lim_{t\to 0} t \cdot x$ exists for all $x \in M_{Hdg}$.

Indeed, these properties are [Sim97, Lemma 10.4, Corollary 10.3].

Then the picture above respects the assumptions of [HR15, Corollary 1.3.3], proving the following theorem.

Theorem 7.2. The inclusions i and j induce isomorphisms of mixed Hodge structures in cohomology

$$H^k(M_{Dol}) \cong H^k(M_{Hdg}) \cong H^k(M_{DR}).$$

We are now left to understand how ψ acts on the mixed Hodge structures. A first step in this direction, is to choose generators of pure weights for the various cohomology rings and track their weight along the various isomorphisms. In order to do this, we take the universal bundle $(\mathcal{E}_{\text{univ}}, \nabla_{\text{univ}})$ on M_{Hdg}.

[**TO DO**: existence of universal objects]

First of all notice that $(\mathcal{E}_{\text{univ}}, \nabla_{\text{univ}})$ restricts to the universal bundles on both M_{Dol} and M_{DR}, namely

$$i^*(\mathcal{E}_{\text{univ}}, \nabla_{\text{univ}}) = (\mathcal{E}_{\text{univ}}, \varphi_{\text{univ}}), \quad j^*(\mathcal{E}_{\text{univ}}, \nabla_{\text{univ}}) = (\mathcal{E}_{\text{univ}}, \nabla_{\text{univ}}).$$

The following theorem is [Mar07, Theorem 3], see also [Mar02; HT04].

Theorem 7.3. Let $c_i(\mathcal{E}_{\text{univ}}) \in H^{2i}(M^d_{n,Dol}(X) \times X; \mathbb{Z})$ be the i-th Chern class of the universal bundle. Write its Künneth decomposition as

$$c_i(\mathcal{E}_{\text{univ}}) = \sum_{j+k=2i} e_{i,j} \otimes x_{i,k}.$$

Then the Künneth components $e_{i,j}$ generate the cohomology ring of $M^d_{n,Dol}(X)$.

Using Equation (5.4), this can be rephrased in our context as follows.

Theorem 7.4. The cohomology ring of $M^0_{n-d,Dol}(\mathcal{X})$ is generated by the Künneth components of the Chern classes of the universal bundle on $M^0_{n-d,Dol}(\mathcal{X}) \times \mathcal{X}$.

Since moreover the cohomology of M_{Dol} carries a pure Hodge structure [**TO DO**: add ref], as clearly does X, the classes $c_i(\mathcal{E}_{\text{univ}})$ have pure weight $2i$, while the Künneth components $e_{i,j}$ have pure weight j.

Consider now

$$j^* \circ (i^*)^{-1} \colon H^*(M_{Dol} \times \mathcal{X}) \to H^*(M_{DR} \times \mathcal{X}).$$

Since i, and j are algebraic maps, this is an isomorphism of pure (mixed) Hodge structures, mapping $c_i(\mathcal{E}_{\text{univ}})$ to $c_i(\mathcal{E}_{\text{univ}})$. Equivalently,

$$j^* \circ (i^*)^{-1} \colon H^*(M_{Dol}) \to H^*(M_{DR})$$

is an isomorphism of pure Hodge structures, mapping the Künneth components $e_{i,j}$ of the Chern classes of the universal bundle on $M_{Dol} \times \mathcal{X}$ to the Künneth components $e_{i,j}$ of the Chern classes on the universal bundle on $M_{DR} \times \mathcal{X}$.

Finally, consider

$$\psi^* \colon H^*(M_{DR} \times \mathcal{X}) \to H^*(M_B \times \mathcal{X}).$$

This induces an isomorphism on the cohomology rings, therefore the Künneth components of $\psi^* c_i(\mathcal{E}_{\text{univ}})$ generate the cohomology ring of M_B. But it is not a morphism of mixed Hodge structures. Nontheless, we have the following.

Theorem 7.5. [She17] Under the isomorphism $\psi^* \colon H^*(M_{DR}) \to H^*(M_B)$, the Künneth components $e_{i,j}$ of the i-th Chern class of the universal bundle $c_i(\mathcal{E}_{\text{univ}})$ are mapped to cohomology classes of weight $2i$.

Shende computed the weights of the tautological classes (corresponding to our $e_{i,j}$'s) for a stack of local system on a topological space, i.e. what in our context would be $\mathcal{M}_B(X)$. We will replicate his argument to work with $\mathcal{M}_B(\mathcal{X})$ and get Theorem 7.5.

8 Shende's argument

In order to compare the Hodge structures of the De Rham and the Betti moduli spaces, we employ Shende's idea presented in [She17]. Shende replaced a topological space, in particular a complex manifold, with a simplicial scheme and use this simplicial decomposition to give a presentation of the Betti moduli space as a simplicial scheme. The main point of the construction is that, while the Betti moduli space doesn't admit a universal local system, since such a local system would not be of algebraic nature, this simplicial replacement does. As we saw in the previous system, universal bundles are the key to comparing the cohomology rings because the Künneth components of their Chern classes are generators.

In what follows, we will first explain in detail the idea behind Shende's construction. Then we will see how it applies to our case, when instead of a complex manifold as studied in [She17] we have a stacky curve.

8.1 A simplicial interpretation of the stack of local systems

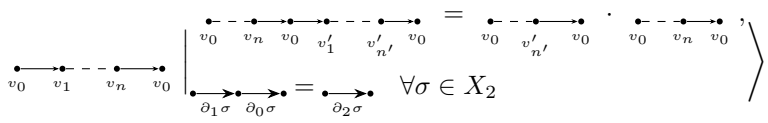
Let X be a complex manifold, choose a base point $v_0 \in X$ for the fundamental group of X. For an algebraic group G, the stack of G-local systems on X is the quotient

$$[\mathrm{Hom}(\pi_1(X, v_0), G)/G], \tag{8.1}$$

where $\pi_1(X, v_0)$ is seen as a discrete algebraic group over $\operatorname{Spec}\mathbb{C}$.

Of course, the definition (8.1) makes sense for any topological space X. By replacing X with a simplicial decomposition, we can replace (8.1) with an equivalent quotient stack.

Notice that we have a simplicial interpretation of the fundamental group. Indeed, if Δ_X is a simplicial set and $v_0 \in \Delta_X^0$ is a vertex, define $E(\Delta_X, v_0)$ the *edge-path group* at v_0 as the group generated by loops of 1-simplices pointed at v_0 under path-composition, modulo the relation killing boundaries of 2-simplices

$E(\Delta_X, v_0) =$ 

Remark 8.1. There is an isomorphism $\pi_1(|\Delta_X|, v_0) \simeq E(\Delta_X, v_0)$.

Proposition 8.2. Let X be a path-connected topological space. Suppose that $\pi_1(X, v_0)$ is finitely presented, e.g. if X is compact. Let Δ_X be a simplicial set whose geometric realization is homotopy equivalent to X, having v_0 as a vertex. Call X_i the i-th skeleton of Δ_X. Then the isomorphism $\pi_1(X, v_0) \simeq E(\Delta_X, v_0)$ induces an equivalence of quotient stacks

$$[\mathrm{Hom}(\pi_1(X, v_0), G)/G] \xrightarrow[\cong]{\mu} \left[\left\{(g_e)_{e\in X_1} \mid g_{\partial_1\sigma}\, g_{\partial_0\sigma}^{-1}\, g_{\partial_2\sigma} = 1 \quad \forall\, \sigma \in X_2\right\}/G^{\#X_0}\right],$$

where $G^{\#X_0}$ acts on $F \coloneqq \left\{(g_e)_{e\in X_1} \mid g_{\partial_1\sigma}\, g_{\partial_0\sigma}^{-1}\, g_{\partial_2\sigma} = 1 \quad \forall\, \sigma \in X_2\right\}$ by edge-boundaries, i.e.

$$(g_v)_{v\in X_0} \cdot (g_e)_{e\in X_1} = (g_{\partial_0 e}\, g_e\, g_{\partial_1 e}^{-1})_{e\in X_1}.$$

Remark 8.3. Both $\mathrm{Hom}(\pi_1(X, v_0), G)$ and F can be expressed as closed subvarieties of some G^N, so indeed the ones described in Proposition 8.2 are well defined algebraic stacks.

Using Remark 8.1, in proving Proposition 8.2 we replace $\pi_1(X, v_0)$ by $E(\Delta_X, v_0)$.

Proof. We actually prove that the quotients prestacks are equivalent. Namely, we show that for S scheme over $\mathbb{C}$ the groupoids

$$G(S) \times \mathrm{Hom}(E(\Delta_X, v_0), G)(S) \rightrightarrows \mathrm{Hom}(E(\Delta_X, v_0), G)(S)$$

and

$$G(S)^{\#X_0} \times F(S) \rightrightarrows F(S)$$

are equivalent, functorially in S. $\square$

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